

DAEN 459 Capstone Senior Design Planning

Fall 2025

Instructor:	Alexander Abuabara	Time:	MW 10:30 – 11:10 (Lecture) F 1:50 – 3:50 (Lab)
Contacts:	abuabara@tamu.edu tamu.zoom.us/my/abuabara	Place:	Zach 211 (Lecture) ETB 1013 (Lab)

Office Hours: MWF 11:30am–1pm or by appointment. I'm usually in my office ETB 4013 during office hours. If you need a different time, I'll do my best to accommodate, please include the course number when reaching out. I'm happy to discuss anything except material covered during an unexcused absence. If other students are already present, feel free to join the conversation. If you would like to discuss a private matter, please schedule an appointment.

Prerequisites: .

Description: First in a two-course sequence for the capstone senior design experience: formation of a senior design team; visitation with the team sponsor; preparation of project groundwork; creation of the project charter; acquisition of an initial data set. Emphasizes ethics, design constraints, applicable standards, project management, technical writing, oral and visual communication.

Objectives: The focus for this course is the project charter and project management packet each team takes on for a client and enhancing each student's communication skills. This is a "C" course, and by university policy, to pass the course you must pass the writing/communications component no matter how high your technical grade is. Students are expected to demonstrate the ABET-based educational outcomes listed below.

Learning Objectives:

1. Translate an open-ended sponsor problem into a concise, data-engineering project charter with measurable goals and constraints.
2. Conduct sponsor meetings, elicit requirements, and document acceptance criteria.
3. Evaluate data-centric solutions for privacy, security, bias, and sustainability impacts; apply IEEE/ACM codes of ethics.
4. Build a work-breakdown structure, Gantt/Agile plans, risk register, and cost estimate for a semester-long technical project.
5. Identify and apply relevant standards (ISO 25012 for data quality, FAIR principles, SEC data-retention, GDPR, etc.).
6. Produce a coherent written proposal and deliver a persuasive 10-minute pitch using professional visuals.

ABET outcomes:

1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. Communicate effectively with a range of audiences.
4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.

Recommended Textbooks:

- *Fundamentals of Data Engineering*. J. Reis, M. Housley (2022)
- *Data Engineering with Python*. P. Crickard (2020)
- *Designing Data-Intensive Applications*. M. Kleppmann (2017)
- *The Data Warehouse Toolkit*. R. Kimball, M. Ross (2013)

Grading Policy:

Grading Component	Weight	Notes
Attendance/Participation	10%	Includes professionalism in sponsor interactions
Peer Review	10%	—
Charter Draft	10%	Formative feedback only
Data Audit Report	10%	Reproducible notebook & memo
Project Management Packet	20%	—
Final Charter (revised)	20%	Signed by sponsor
Oral Pitch & Visuals	20%	Presentation + slide deck

Grades will be calculated on the basis of total points earned. The thresholds for letter grades are: A $\geq$ 90%, B $\geq$ 80%, C $\geq$ 70%, D $\geq$ 60%, F < 60% If deemed necessary, the grade thresholds may be curved at the instructor's discretion.

Attendance is mandatory for sponsor meetings and presentation days. You are expected to attend all class lectures except for university excused absences. The university rule regarding excused absences can be found at <https://student-rules.tamu.edu/rule07>.

Initial instructions for the charter and project management packet are provided below. Additional guidance will be given in class.

Late work will not be accepted except for approved university excused absences. Any disagreements regarding a grade received on any graded material must be discussed within one week of the return of the graded material. No grade will be changed beyond this limit. If you request a regrade, the entire submission will be regarded (i.e. if we made a mistake in your favor, this will also be corrected).

Academic Integrity: Don't cheat! If your temptation is to not turn in work that isn't yours, don't do it. Come see me instead and ask questions, I'm here to help. Content is cumulative, so the earlier, the better. Modeling, coding, data analysis are skills that will open many doors for you: start learning now and set yourself up for success! In keeping with the Texas A&M Student Honor Code, all materials are scanned with plagiarism checkers.

Academic Honesty: The Industrial & Systems Engineering Faculty uphold the Aggie Honor Code as an axiom of academic excellence. We consider its sincere observance to be essential for membership in our department and Texas A&M. We extend you the trust conferred to those who faithfully adhere to our honor code. Abuse of this trust is intolerable, thus we will report and assign an extreme penalty to those who do not stand with us in preserving the integrity symbolized by the Aggie Honor Code: **"An Aggie does not lie, cheat, or steal or tolerate those who do."** We will request the Aggie Honor Office to assign a grade of F for any violation of the Aggie Honor Code, no matter how minimal the violation may be. The Honor Council Rules and Procedures can be found at <https://aggiehonor.tamu.edu>

The Americans with Disabilities Act (ADA): A federal anti-discrimination statute that provides comprehensive civil rights protection for persons with disabilities. Among other things, this legislation requires that all students with disabilities be guaranteed a learning environment that provides for reasonable accommodation of their disabilities. If you believe you have a disability requiring an accommodation, please contact Disability Services, currently located in the Disability Services building at the Student Services at White Creek complex on west campus or call 979-845-1637. For additional information visit <https://disability.tamu.edu>

Statement on Diversity and Inclusion: It is important to us that all students feel accepted and comfortable in this class. If for any reason you feel disrespected, uncomfortable, or unsafe in our classroom please discuss your concerns with us via the medium you are most comfortable with. If you do not feel comfortable communicating directly to us, please reach out to Dr. Ntaimo (department head).

Additional Information: All information related to the course, including handouts, will be made available on <https://canvas.tamu.edu>. All communication related to the course will be sent electronically via Canvas as announcements or delivered to your @tamu.edu mailbox. You are expected to check your email regularly.

Some details about the course and assignments:

- This **capstone course** serves as an integrative, high-impact educational experience that challenges students to synthesize their accumulated knowledge and apply it to a complex, real-world problem. The **capstone projects** encourage students to adopt a hands-on approach by building solutions from the ground up rather than relying only on existing tools or templates. The **goal** is to improve essential skills such as project management, communication, and interdisciplinary analysis, thereby preparing students for professional or academic success. Figure 1 provides a comprehensive overview of how data moves through a system, highlighting the sequential steps of data engineering and the critical underlying disciplines that ensure its success.

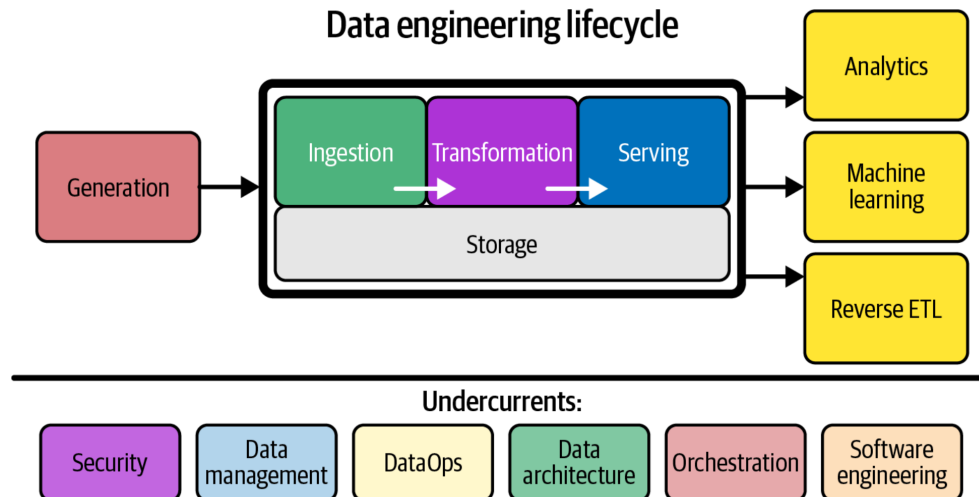


Figure 1: Components and undercurrents of the data engineering lifecycle

Image extracted from *Fundamentals of Data Engineering: Plan and Build Robust Data Systems*, J. Reis, M. Housley, 2022.

Unlike *busy work* which often involves repetitive, low-cognitive tasks with limited connection to meaningful learning outcomes, this capstone project demands: critical thinking, research, and collaboration. Here “reinventing the wheel” is not *busy work* but a deliberate method for gaining insights, building confidence, and preparing for real-world innovation. Drawing inspiration from [Richard Feynman's](#) quote, “What I cannot create, I do not understand”, these are some of the great reasons to *reinvent the wheel*:

- Learn how wheels are made
 - Teach others about wheels
 - Learn about the inventors of wheels
 - Be able to change wheels (or fix) when they break
 - Build a better wheel (for some definition of better)
 - Help someone in need of a very special wheel (maybe for a wheelchair?)
 - Learn a tiny slice of what it means to build a larger system (such as a vehicle)
- The **V-model** (Fig. 2) provides a structured framework ideal for planning the Data Engineering Capstone Senior Design courses. It guides students from defining project goals and requirements to detailed system design and implementation. Each phase on the left is mirrored by corresponding testing and validation on the right, ensuring alignment between design and outcome. This model promotes a detailed and careful planning, iterative development, and continuous verification, helping students understand the full lifecycle of a data engineering system, from concept and design to deployment and maintenance. A more detailed model will be explored later.
 - The **Project Management (PM) Packet** is the single, living document that guides a data-engineering capstone team from idea approval through final hand-off. Think of it as the project's constitution: it captures what you will build, how, by whom, and when; so faculty, sponsors, and teammates always have one authoritative source of truth. Keep the packet lean (about 10 pages), version-controlled (in Git), and revisit it whenever scope, risks, or constraints change. A well-maintained PM Packet not only streamlines your semester, it showcases professional project-management discipline to future employers. How it's used:
 1. **Proposal phase:** Submitted with your charter to secure faculty and sponsor sign-off.
 2. **Execution phase:** Updated after each sprint review; serves as the agenda for weekly stand-ups.
 3. **Evaluation phase:** Forms the backbone of your final report and memo, proving objectives were met.

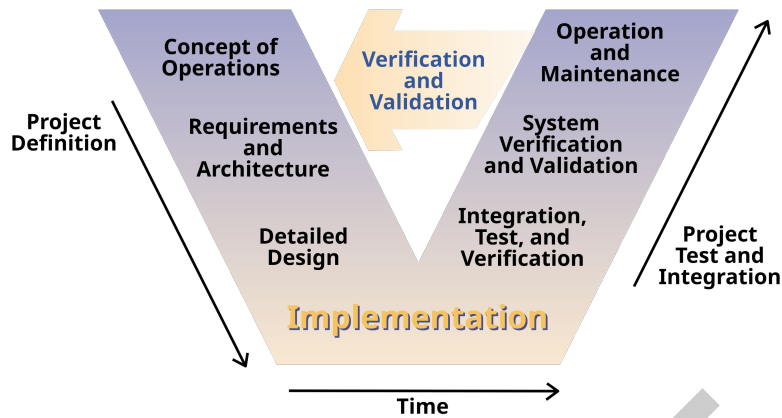


Figure 2: Systems Engineering Process

Image extracted from *Clarus: Concept of Operations*, Federal Highway Administration (FHWA), 2005.
(Publication No. FHWA-JPO-05-072, authors: L. Osborne, J. Brummond, R. Hart, M. Zarean, S. Conger.)

PM Packet Element	Purpose
Work Description	Plain-language summary of what the team will build or analyse.
Deliverables	Defines concrete outputs (code, datasets, report, poster) to hand over.
Project Boundaries	States what is <i>in</i> and <i>out</i> to guard against scope creep.
Acceptance Criteria	Minimum conditions the sponsor must see to sign off.
Constraints	Non-negotiable limits (tools, data privacy, semester length, etc.).
Assumptions	Conditions taken as true (e.g. data access) that may later change.
Quality and Performance Metrics	How accuracy, reliability, and efficiency will be judged.
Metrics & Verification Methods	How each metric will actually be measured, tested, and logged.
Timeline and Milestones	Semester checkpoints that trigger progress reviews and pivots.
Task Priority	Ranks tasks so critical-path items are tackled first.
Risks & Risk Management Plans	Lists top pitfalls and the team's mitigation / contingency moves.
Cost Estimates and Budgeting	Forecasts hours, cloud credits, and purchases to stay on budget.
Resources	People, software, hardware, and data the project will rely on.

- The **Project Charter** is a concise, authoritative document that formally authorizes a project. In a two-semester capstone sequence, the *final sponsor-signed charter* is the *green light* to proceed to implementation in the second capstone semester. Issued or endorsed by the project sponsor, it grants the project manager and team the authority to commit organizational resources and establishes a common understanding of the project's purpose, scope, and constraints. Treat the charter as both a contract and a living reference; if conditions change, revise it formally rather than drifting informally. An outline of the charter elements is presented below. Generally speaking, these are important aspects that need to be addressed in the charter:
 - Authority & accountability:** Sponsor sign-off clarifies who owns vision and acceptance.
 - Unified vision:** Forces the team to articulate a shared understanding before deep technical work.
 - Baseline for change control:** Scope or resource changes must be justified against the charter, avoiding unpleasant surprises.
 - Early risk exposure:** Listing assumptions and constraints surfaces ethical, legal, or technical landmines while there is time to plan.
 - Communication tool:** Because it is short, executives and non-technical stakeholders will actually read it.

Charter Element	Purpose
Background / Business Need	Link the project to strategic goals or a sponsor pain point.
Problem / Opportunity Statement	Define the specific gap, risk, or opportunity the team will address.
Project Objectives & Success Criteria	List SMART goals and measurable indicators of success.
Scope In / Scope Out	Establish boundaries to prevent mission creep and side quests.
High-Level Requirements	Capture essential functional or data requirements.
Stakeholders & Roles	Identify sponsor, prj. manager, team, users, regulators, authority lines.
Major Milestones & Schedule	Provide a high-level timeline to show pacing.
Budget / Resource Commitments	State the dollar or effort limits and promised assets.
Assumptions & Constraints	Surface conditions accepted as true and limits to respect.
Risks & Mitigations	Highlight top threats and first-pass countermeasures.

Course Outline: (tentative)

Week	Lecture (Tue/Thu)	Lab (Fri)	Milestones & Deliverables
1	Introduction; capstone overview; team formation mixer	Data-engineering project exemplars	Team roster + expertise/skills
2	Human-centered problem framing; sponsor needs template	Ethics in data acquisition & use	—
3	First sponsor visitation (onsite / virtual)	Guest panel: industry sponsors	Visit-notes memo
4	Writing the project charter (scope, objectives, KPIs)	Standards & regulations landscape	Charter draft 1
5	Data discovery: sourcing, licensing, privacy screening	FAIR & open-data principles	—
6	Preliminary data acquisition	Data quality metrics (ISO 25012)	Initial data audit report
7	Design constraints (computational, ethical, economic, environmental)	Mid-semester reflection	—
8	Project management tools (Git, Jira, CI/CD for data pipelines)	Agile vs. waterfall for capstones	Work breakdown structure (WBS) + timeline
9	Risk analysis workshop (Monte-Carlo in Python)	"When data projects fail" (guest)	List of potential project risks (short report)
10	Security & privacy threat modeling (STRIDE, DP taxonomy)	Legal counsel Q&A (GDPR, HIPAA)	—
11	Data governance & documentation standards	Writing technical specs	Data governance plan
12	Budgeting & resource planning (cost calculators)	Project finance primer	—
13	Crafting executive summaries & elevator pitches	Slide-design lab (storytelling)	Charter final + pitch deck draft
14	Rehearsal presentations with peer review and invited professors (/advisors)	Design showcase logistics	—
15	Final charter submission & pitch to sponsors	Course wrap-up; next-semester preview	Sponsor-signed charter + final pitch